

SynthProof — Making a DP Synthetic-Data Release Checkable

The field agrees on **what** to disclose about a differentially private release, and knows the ways it leaks. What no one ships is a release a stranger can **verify from the artifact alone**. SynthProof builds that: a signed, machine-checkable privacy label with a 14-check release-boundary auditor — implementing the label Dibia et al. propose and filling the two gaps they leave (a signature, and honest reporting of the audit's limits).

Position: integration, not invention — and stated as such throughout

The problem, quantified

Not a hypothetical — measured on real releases.

0 / 290

HuggingFace synthetic-data datasets declare DP with an ϵ (refreshed 2026-09-17)

0 / 12

flagship real DP deployments are signed or machine-checkable (Apple, Google, US Census...)

blind trust

practitioners do NOT verify DP guarantees — they trust implicitly (Song, CSCW'24)

Our contribution — three pillars

1 Release-boundary auditor

RB1–RB14. Reads only the published artifact — never the data or the producer's code — and flags every channel that leaks *outside* ϵ : the run seed, exact row count, unkeyed fingerprint, unlabelled evaluation, uncharged domain/discretisation, undisclosed amplification, and multi-table (foreign-key degree, join cardinality, cross-table linkage).

2 Signed, checkable label

The Privacy Data Sheet as a Croissant 1.1 extension: Dibia's 9 categories + **an Ed25519 signature + an operating-range field** (the two gaps Dibia leaves). Conformance = passing the auditor. Validated against the official MLCommons validator with 0 warnings.

3 Gated end-to-end pipeline

One reproducible command: synthesise → sign → emit Croissant → assert RB1–RB14 clean, conformant, signature verifies, MLCommons-validates → one report with hashes. The runnable artifact an examiner can execute.

Honest novelty (checked, research/25): the leak *phenomena* are known and empirical MIA/GDP auditing is crowded — so we claim neither. What is **not** built elsewhere is the **signed, artifact-only, machine-checkable checker**: Dibia proposes the label and builds no checker, no signature, no limit-reporting. That is our engineering contribution.

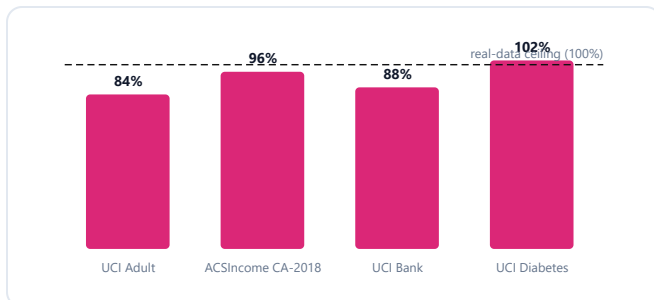
How we differ from every paper — the minute differences

All 11 papers are Claude-verified deep reads (research/18). Red = the exact gap they leave; green = what SynthProof adds.

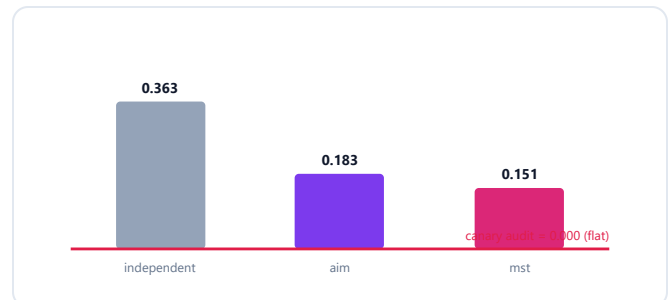
#	Paper	What they do	The gap	What SynthProof adds
P1	Stadler — Groundhog Day USENIX'22	Proved synthetic data leaks membership for outliers; showed DP impls read the domain from data.	Runs shadow-model attacks; ships no signed/checkable artifact.	We turn that leak into a checkable field (RB6 domain_source) in a signed label.
P2	Annamalai — Tight Auditing USENIX'24	Audits the mechanism; black-box audits read $\epsilon \approx 0$ even at true $\epsilon = 4$.	Needs the code / worst-case; no document-level audit.	We cite it as the honesty bound for audited $\epsilon = 0$; our GDP audit (research/24) replicates it.
P3	Cebere — Grey-Box Auditing PoPETs'26	Found 13 privacy bugs in 12 DP libraries.	Audits library code, not the shipped release.	We audit the released artifact; same neighbours+fixed-randomness idea powers our seed-replay (RB1).
P4	Ganev — Domain Extraction ICLR'25	Reading a column's min/max from private data breaks end-to-end DP.	No artifact field to declare how the domain was obtained.	RB6 domain_source makes it a checkable value (declared/codebook/charged vs inferred).
P5	Ganev — Discretization CCS'25	Data-derived bins (quantile/k-means) leak the input distribution.	No checkable release field for the binning basis.	RB9 discretization_source flags data-derived bins as a leak.
P6	Ganev — SMOTE and Mirrors ICLR'26	Naive utility metrics miss real leaks; you need MIA.	Diagnoses the metric gap; builds no release check.	Our sanity-gate + MIA arm act on exactly this; a bare 'no leak' is never trusted.
P7	Mohapatra — Missing Data VLDB'24	Studies whether missingness hurts utility / helps privacy (Adult + Bank).	Never audits membership; no amplification disclosure.	RB10 amplification_disclosure; base for our bounded imputation audit.
P8	McKenna — AIM VLDB'22	State-of-the-art marginal DP synthesizer (we USE it).	Ships a model, no signed/boundary check.	We integrate AIM + MST, audit them, and add AIM's adaptive budget (research/26).
P9	Abowd — Census TopDown HDSR'22	Real deployment; publishes invariants outside ϵ .	Invariants documented in prose only, not machine-checkable.	RB8 public_invariants + RB11-14 relational make them a checkable field.
P10	Dibia — DP Privacy Label PoPETs'26	Expert 9-category privacy label — our base paper.	No signing, no measurement-limit reporting ('privacy theater'); builds no checker.	★ We BUILD it: Ed25519 signature + operating-range field + boundary-audit conformance.
P11	Song — Mental Models CSCW'24	Practitioners do NOT verify DP guarantees — they trust blindly.	Diagnoses the problem; builds no tool.	★ The premise for an automatic checker a third party can run.

Results — where we are better

Downstream ML utility: OUR mechanisms (AIM/MST) reach the real-data ceiling on all 4 datasets



GDP audit recovers a real privacy number where the canary audit collapses to 0



UCI Adult

best structure **aim** @ $\epsilon=8 \rightarrow 0.0105$ err

CENSUS

ACSIncome CA-2018

best structure **pairwise** @ $\epsilon=8 \rightarrow 0.0202$ err

CENSUS

best ML utility **mst** @ $\epsilon=2 \rightarrow 84\%$ of real-data F1

best ML utility **mst** @ $\epsilon=2 \rightarrow 96\%$ of real-data F1

UCI Bank Marketing

FINANCE

best structure **pairwise** @ $\epsilon=8 \rightarrow 0.0289$ err

best ML utility **mst** @ $\epsilon=1 \rightarrow 88\%$ of real-data F1

UCI Diabetes 130

HEALTHCARE

best structure **mst** @ $\epsilon=0.5 \rightarrow 0.1052$ err

best ML utility **mst** @ $\epsilon=1 \rightarrow 102\%$ of real-data F1

Across **4 datasets** — **census ×2, finance, healthcare** (the one healthcare table, Diabetes 130, added so the medical framing is backed by a real clinical dataset): our select-measure mechanisms take the best downstream utility everywhere, reaching up to **96% of real-data F1**. On healthcare they also cut correlation error $\sim 4\times$ below the baselines. The empirical GDP audit gives $\mu = \text{independent } 0.363 \cdot \text{aim } 0.183 \cdot \text{mst } 0.151$ — informative where the black-box canary audit reads a flat **0.000**.

What we deliberately do NOT claim

- **Not "15/15 parameters pass"** — that figure does not replicate; we report the RB1–RB14 checks and the audit ceiling instead.
- **Not "we recover the original data"** — we detect what a release *leaks beyond its ϵ* (e.g. a published seed replays membership 15/15). We never reconstruct private records.
- **Audited $\epsilon = 0$ is instrument-limited**, not proof of no leakage — the honest headline, and exactly why the operating-range field exists.
- **Multi-table auditing ships; multi-table synthesis does not** — no validated relational generator yet (declared future work / Paper 2).
- **The generators (AIM/MST) are integrations** of published methods, not novel mechanisms.

One line for the paper

The field has the guarantee parameters, the leak phenomena, and empirical attacks; what it lacks — and what Dibia et al. (2025) call for but do not build — is a **signed, machine-checkable release label a third party can verify from the artifact alone**. We build it, fill its two gaps, and show every prominent real release fails it today.

Generated by `scripts/gen_contribution_report.py` from committed results (H1 grids, `results/gdp_audit*.json`, `research/wild_audit/`). Per-paper rows quoted from `research/18` (Claude-verified). No number invented; honest limits above are part of the report, not a disclaimer.